CWS- Offshore Bat Monitoring Project

Manual Verification Data Summary

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1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on treaty lands as well as the traditional and unceded territories of First Nations, Inuit, and Métis Peoples across all regions of Canada, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

Figure 1: Offshore wind turbine installed about 27 miles off the coast of Virginia Beach by the Coastal Virginia Offshore Wind pilot. Photo by Jess Stormberg/BOEM

2 Background

There is a growing need to understand bat activity in offshore environments as the exploration and development of offshore wind energy facilities expands. Several bat species are known to use offshore routes during migration and movement, including three Canadian migratory species, Hoary Bat (*Lasiurus cinereus)*, Eastern Red Bat (*Lasiurus borealis)*, and Silver-haired bat (*Lasionycteris noctivagans)*, which have been assessed as Endangered by COSEWIC (Solick and Newman 2021; Canada and Change 2024).

To better understand bat use of offshore environments and assess the potential impacts of offshore wind energy facilities, the Canadian Wildlife Service is monitoring bats offshore using both stationary and mobile acoustic detectors, following the guidelines set forth by the Regional Wildlife Science Collaborative for Offshore Wind (RWSC, n.d.).

Figure 2: Silver-haired bat (*Lasyonicteris noctivagans*). Photo by Jason Headley

3 Methods

The Canadian Wildlife Service deployed acoustic units to record bats at 13 stationary sites along the maritime coastline, and 5 mobile transects on 5 different vessels (Cartier, DFO-Discovery, NRCAN-Discovery, Coriolis, and James Cook). The NRCAN-Discovery vessel carried 3 microphones positioned on different sections of the ship (main mast, met mast A, and met mast B). The James Cook vessel had two microphones positioned on different sections of the ship (main and met mast). Sampling occurred mainly in the summer and fall of 2025, with additional sampling from 2024 and 2026 submitted for review.

3.1 Data processing

Full-spectrum recordings from the sampling periods were processed using two automatic classifiers: Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Northeastern North America classifier. Manual identification efforts focused on 7 species: Big Brown Bats (*Eptesicus fuscus*), Silver-haired Bats (*Lasionycteris noctivagans*), Hoary Bat (*Lasiurus cinereus*), Eastern Red Bat

(*Lasiurus borealis*) Little Brown Myotis (*Myotis lucifugus*), Northen Myotis (*Myotis spententrionalis*) and Tri-colored bat (*Perimyotis subflavus*).

The analysis workflow followed processing standards established by the North American Bat Monitoring Program (NABat) (Reichert et al. 2018). Only recordings that were identified as a bat by Kaleidoscope were manually verified. All recordings with automated classifications underwent complete manual verification. Species identifications were validated using reference call parameters described by Szewczak (2022) and Vesper (2022) , in accordance with NABat manual vetting protocols (Reichert et al. 2018).

4 Results and Discussion

In total, 234,137 recordings were submitted to SENSR for review. Of these, 211,376 were identified as Noise by Kaleidoscope, and the remaining 22,761 were manually reviewed and assigned a species or species-grouping ID. Manual verification flagged an additional 2,432 recordings as noise that Kaleidoscope had erroneously identified as bats, leaving a total of 20,335 bat recordings across all sites. Of these, 15 were recorded along the mobile ship-based routes, while the remainder came from stationary acoustic recorders deployed along the coastline (Figure 3, Appendix B).

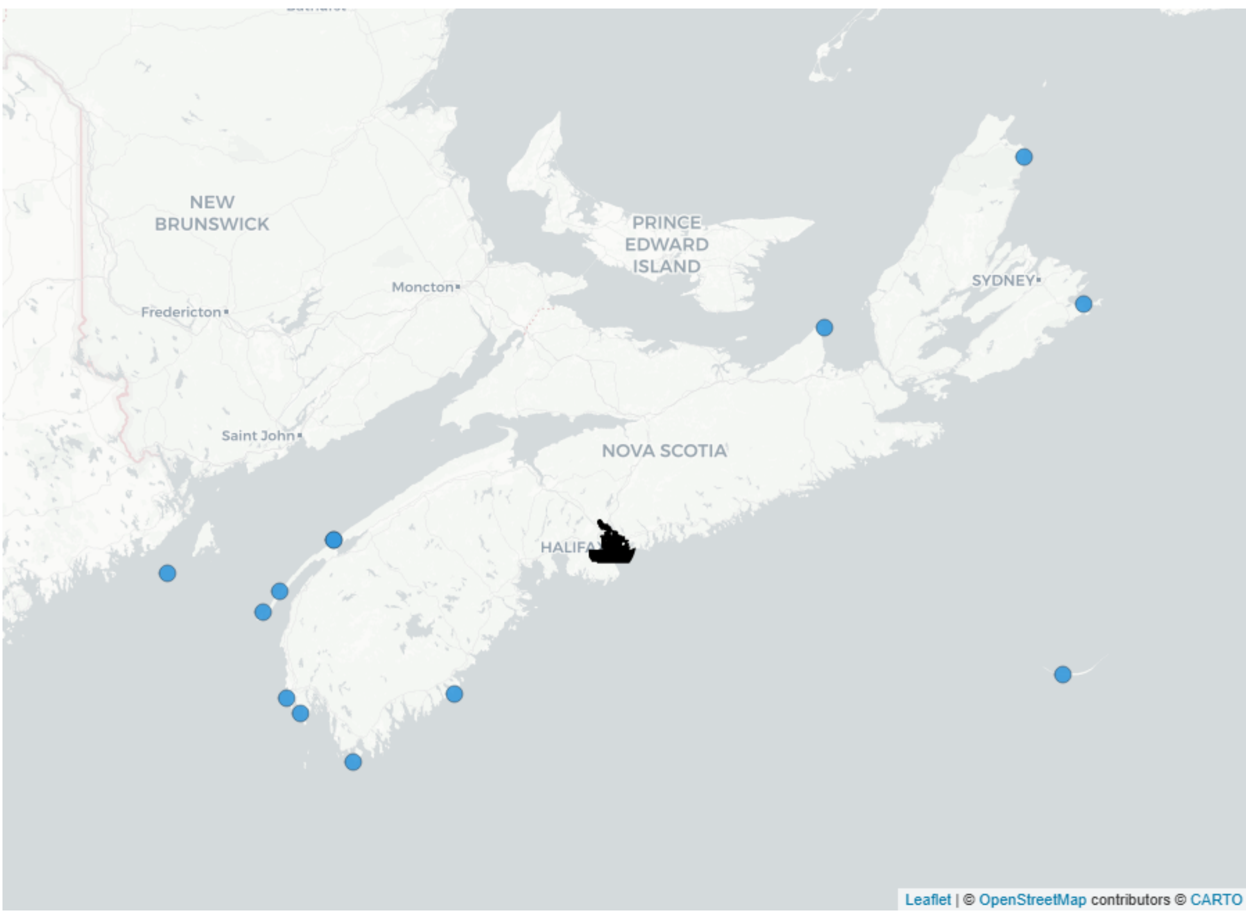


Figure 3: Acoustic bat monitoring sites across the Nova Scotia study area. Stationary recording stations are shown as circular markers and mobile (vessel-based) transects as boat icons.

All species were recorded across the project sites, but detections varied among sites and across the survey months. The most commonly recorded species was the hoary bat (*Lasiurus cinereus*), with a total of 519 detections, followed closely by the silver-haired bat (*Lasionycteris noctivagans*) with 488 and the eastern red bat (*Lasiurus borealis*) with 235. The greatest overall bat activity was recorded at Prim Point Lighthouse, which had its highest activity in June 2025 (Figure 3, Appendix B).

Feeding buzzes were recorded at six of the monitored sites, providing direct evidence of foraging activity rather than commuting flight alone, though this foraging may have been opportunistic rather than reflecting habitual use of these areas. The greatest number of feeding buzzes was recorded at Brier Island (166), followed by Pinkney Point (90), with the remaining four sites contributing smaller numbers. These feeding buzzes originated primarily from recordings identified as silver-haired bats, indicating that this species was actively foraging within the study area. No social calls were detected at any site over the course of the survey.

Overall, the quality of recordings was good. Noise files were numerous across all sites, which was expected based on similar studies and is likely due to the proximity of recording equipment to the ships and lighthouses. We recommend that future sampling continue to follow the Regional Wildlife Science Collaborative for Offshore Wind guidance on sampling techniques.

5 Appendix A

Species codes and their definitions

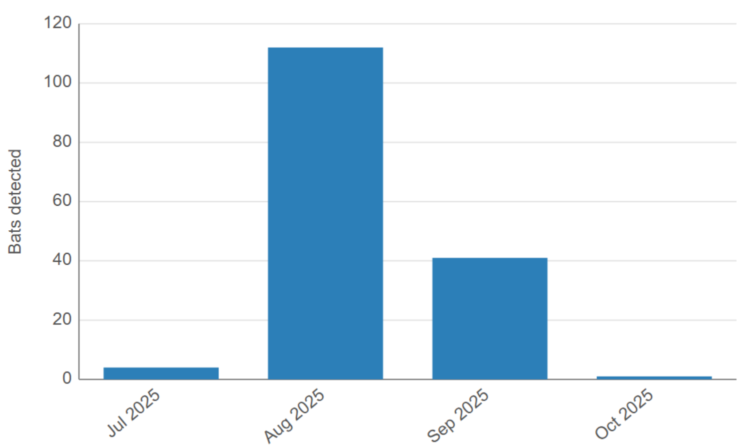
CommonName	ScientificName	Code	Definition
Big Brown Bat	Eptesicus fuscus	EPFU	Calls that have diagnostic features identifying it as Eptesicus fuscus
Big Brown Bat / Silver-haired Bat	Eptesicus fuscus / Lasionycteris noctivagans	EPFULANO	Calls that could be attributed to either Eptesicus fuscus or Lasionycteris noctivagans
Eastern Red Bat	Lasiurus borealis	LABO	Calls that have diagnostic features identifying it as Lasiurus borealis
Eastern Red Bat / Little Brown Myotis	Lasiurus borealis / Myotis lucifugus	LABOMYLU	Calls that could be attributed to either Lasiurus borealis or Myotis lucifugus
Eastern Red Bat / Tricolored Bat	Lasiurus borealis / Perimyotis subflavus	LABOPESU	Calls that could be attributed to either Lasiurus borealis or Perimyotis subflavus
Hoary Bat	Lasiurus cinereus	LACI	Calls that have diagnostic features identifying it as Lasiurus cinereus
Hoary Bat / Silver-haired Bat	Lasiurus cinereus / Lasionycteris noctivagans	LACILANO	Calls that could be attributed to either Lasiurus cinereus or Lasionycteris noctivagans
Silver-haired Bat	Lasionycteris noctivagans	LANO	Calls that have diagnostic features identifying it as Lasionycteris noctivagans
Little Brown Myotis	Myotis lucifugus	MYLU	Calls that have diagnostic features identifying it as Myotis lucifugus
Little Brown Myotis / Northern Myotis	Myotis lucifugus / Myotis septentrionalis	MYLUMYSE	Calls that could be attributed to either Myotis lucifugus or Myotis septentrionalis
Northern Myotis	Myotis septentrionalis	MYSE	Calls that have diagnostic features identifying it as Myotis septentrionalis
Tricolored Bat	Perimyotis subflavus	PESU	Calls that have diagnostic features identifying it as Perimyotis subflavus
40kHz Frequency Myotis		40kMyo	Various species of Myotis that have a characteristic frequency in the range of 35-40kHz
High Frequency Bat		HIGH FREQUENCY	Various species with pulses having a characteristic frequency higher than ~35kHz
Low Frequency Bat		LOW FREQUENCY	Various species with pulses having a characteristic frequency lower than ~30kHz
Unknown Bat		NoID	Bat calls but no grouping category applies
Feeding buzz		FB	Feeding buzz present in one or more of the recordings
Social call		SC	Social call present in one or more of the recordings
Noise		Noise	No bat recorded

6 Appendix B

Species detection summaries for each monitoring site.

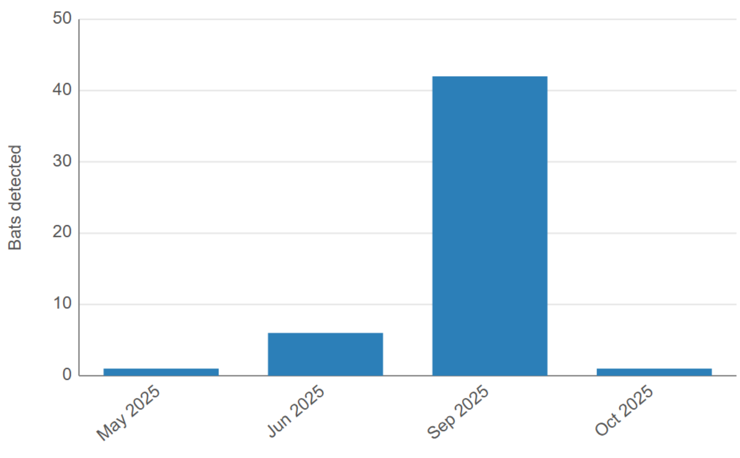
Coastline-KejiSeaside

Species Code	Total Detected
EPFU	3
LABO	5
LABOMYLU	30
LABOPESU	8
LACILANO	4
LANO	8
MYLU	80
MYLUMYSE	4
MYSE	4
PESU	9
High Frequency	3
Noise	4,139
Total Bats	**158**



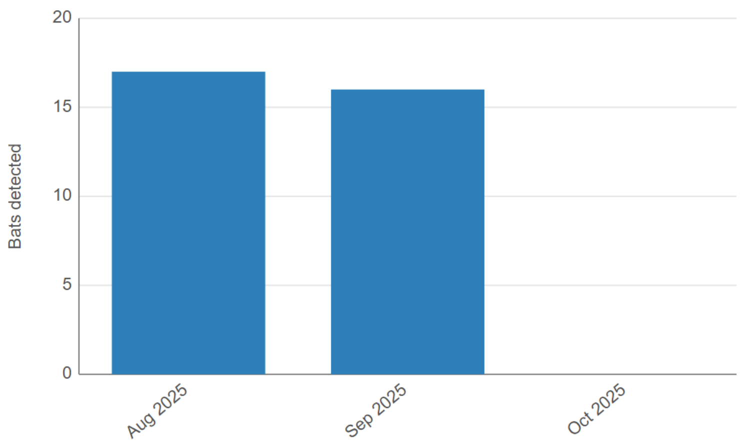
Coastline_HartlenPoint

Species Code	Total Detected
EPFULANO	1
LABO	1
LABOMYLU	13
LACI	7
LACILANO	2
LANO	9
MYLU	7
PESU	1
High Frequency	9
Noise	700
Total Bats	**50**



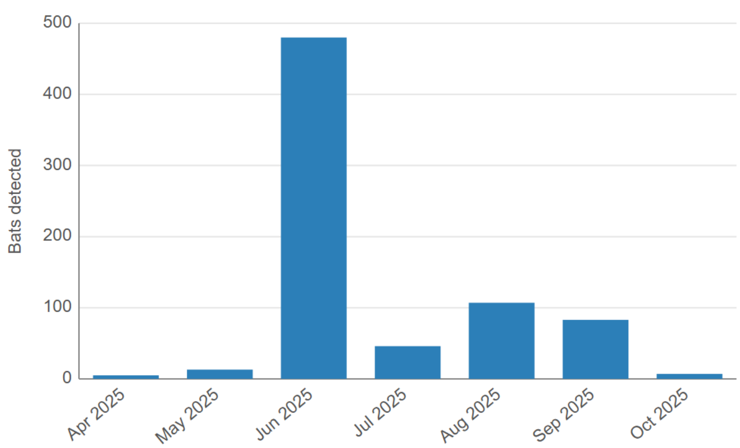
Coastline_NeilsHarbour

Species Code	Total Detected
LABO	5
LABOMYLU	3
LABOPESU	3
LACI	6
LACILANO	3
MYLU	12
High Frequency	1
Noise	1,442
Total Bats	**33**



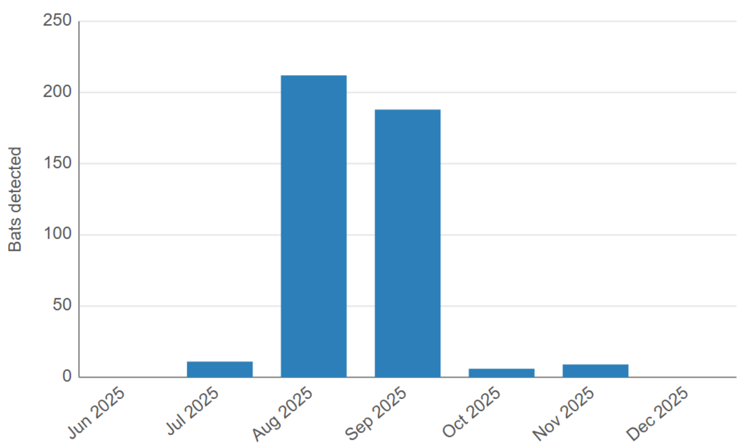
Lighthouse_PrimPoint

Species Code	Total Detected
EPFU	6
EPFULANO	22
LABO	22
LABOMYLU	10
LABOPESU	21
LACI	441
LACILANO	52
LANO	61
MYLU	28
MYLUMYSE	1
MYSE	3
PESU	1
40KMYO	28
High Frequency	13
Low Frequency	9
NoID	23
Noise	965
Total Bats	**741**



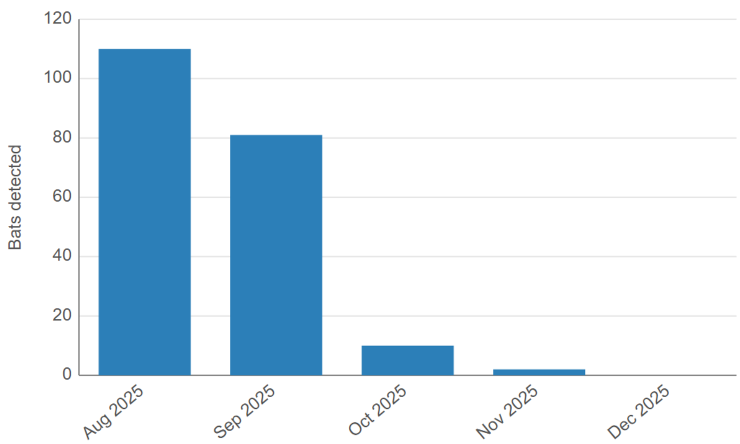
NB-GOM-ISL-O-MachiasSealIsland

Species Code	Total Detected
EPFU	6
LABO	65
LACI	10
LANO	191
MYLU	16
PESU	5
High Frequency	5
Low Frequency	128
FB	42
Noise	8,066
Total Bats	**426**



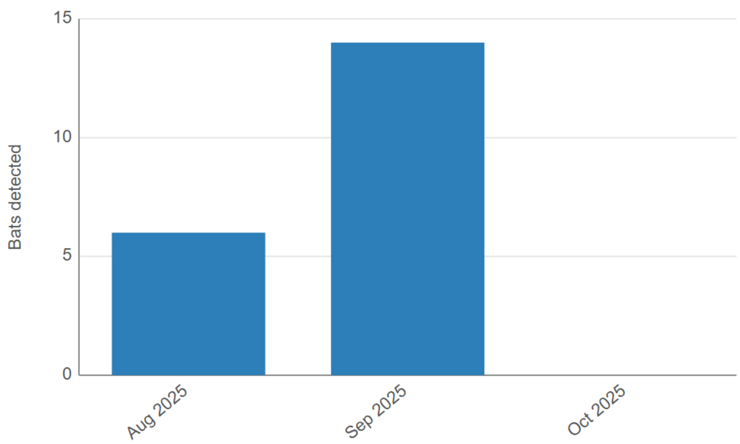
NS-BOF-ISL-Brier

Species Code	Total Detected
EPFU	3
LABO	25
LACI	23
LANO	62
High Frequency	3
Low Frequency	87
FB	181
Noise	43,594
Total Bats	**203**



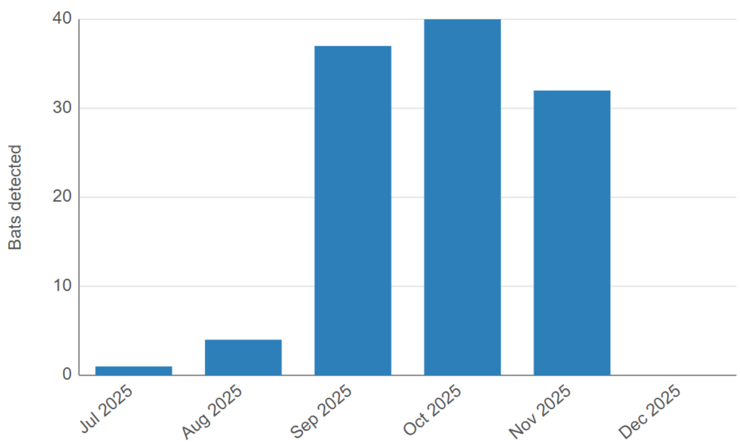
NS-CBH-CST-CapeGeorge

Species Code	Total Detected
LABO	13
LANO	1
PESU	1
High Frequency	4
Low Frequency	1
Noise	24,746
Total Bats	**20**



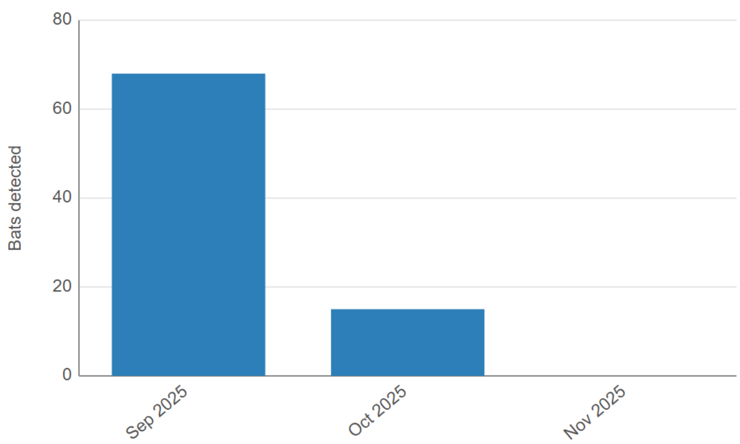
NS-ESS-ISL-O-SableIsland

Species Code	Total Detected
EPFU	2
LABO	33
LACI	1
LANO	29
High Frequency	2
Low Frequency	47
FB	4
Noise	6,346
Total Bats	**114**



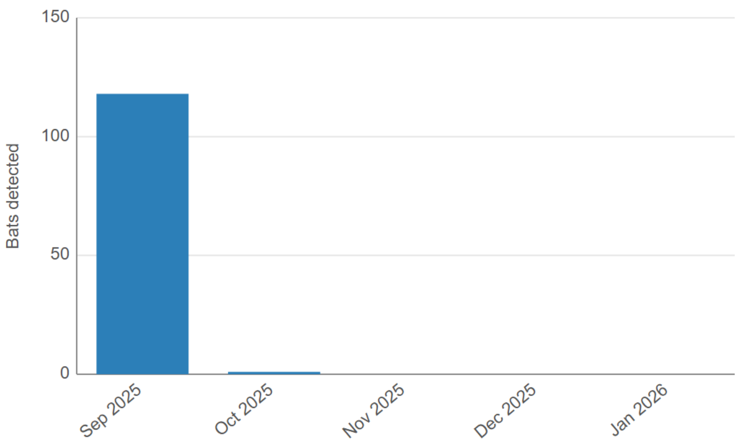
NS-LOB-CST-CapeForchu

Species Code	Total Detected
LABO	30
LACI	1
LANO	15
MYSE	1
PESU	2
High Frequency	9
Low Frequency	25
FB	1
Noise	23,214
Total Bats	**83**



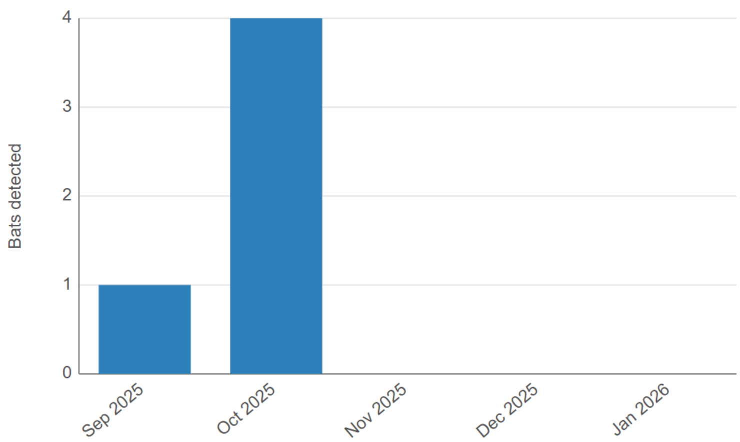
NS-LOB-CST-PinkneyPoint

Species Code	Total Detected
LABO	2
LANO	67
Low Frequency	50
FB	90
Noise	2,609
Total Bats	**119**



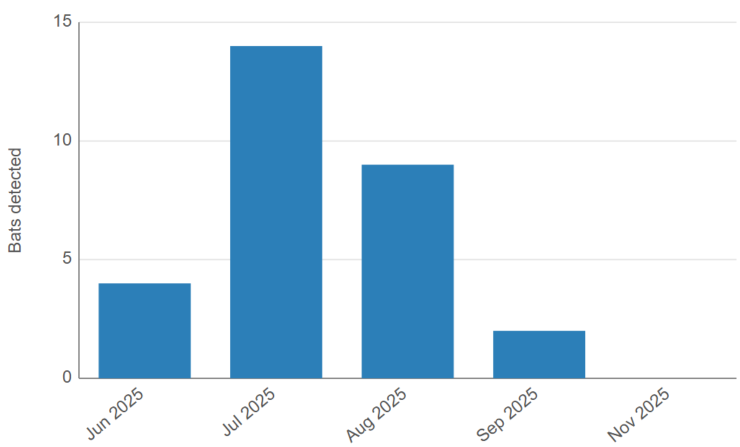
NS-LOB-CST-SwimsPoint

Species Code	Total Detected
LABO	2
High Frequency	1
Low Frequency	2
Noise	2,077
Total Bats	**5**



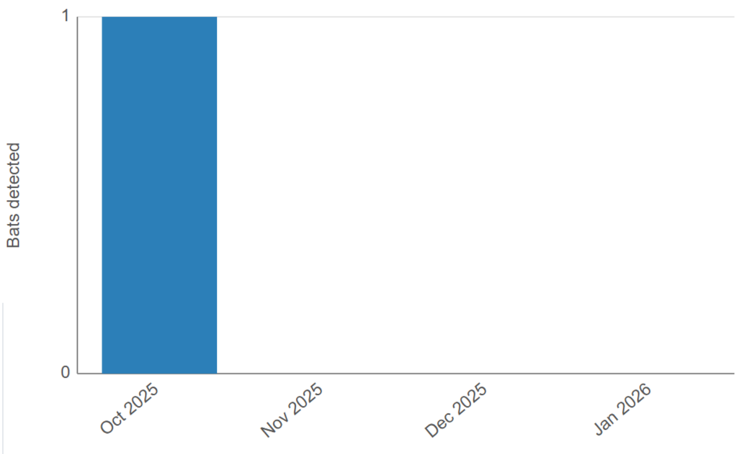
NS-OBF-CST-BoarsHead

Species Code	Total Detected
EPFU	1
LABO	10
LACI	2
LANO	3
MYLU	2
High Frequency	2
Low Frequency	9
Noise	790
Total Bats	**29**



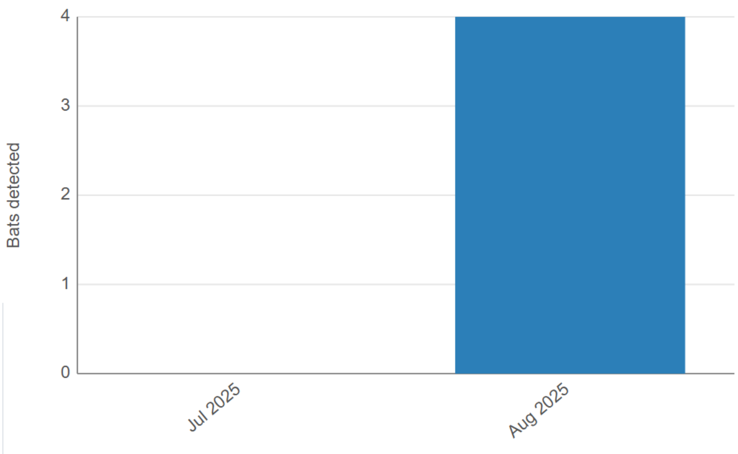
NS-SCA-CST-MainaDieu

Species Code	Total Detected
LABO	1
FB	1
Noise	6,751
Total Bats	**1**



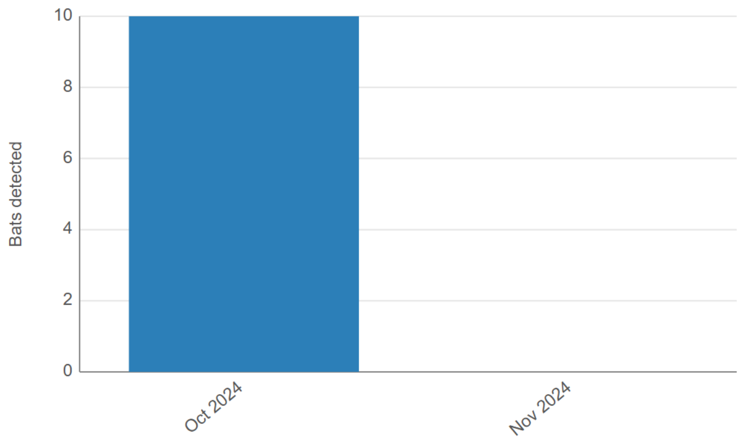
DFO-Cartier

Species Code	Total Detected
LABOPESU	2
PESU	2
Noise	4,734
Total Bats	**4**



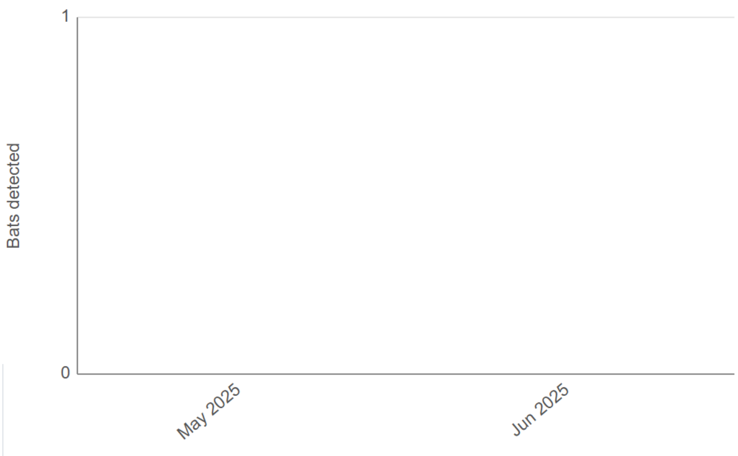
DFO-Discovery

Species Code	Total Detected
EPFULANO	8
Low Frequency	2
Noise	38,916
Total Bats	**10**



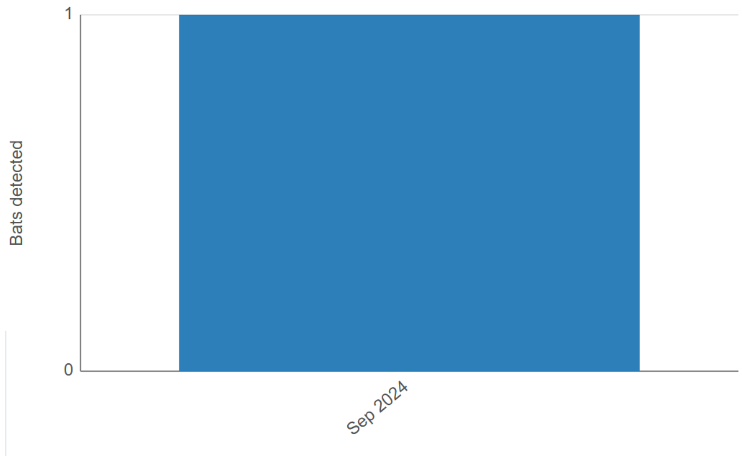
NRCAN-Coriolis

Species Code	Total Detected
Noise	13,284
Total Bats	**0**



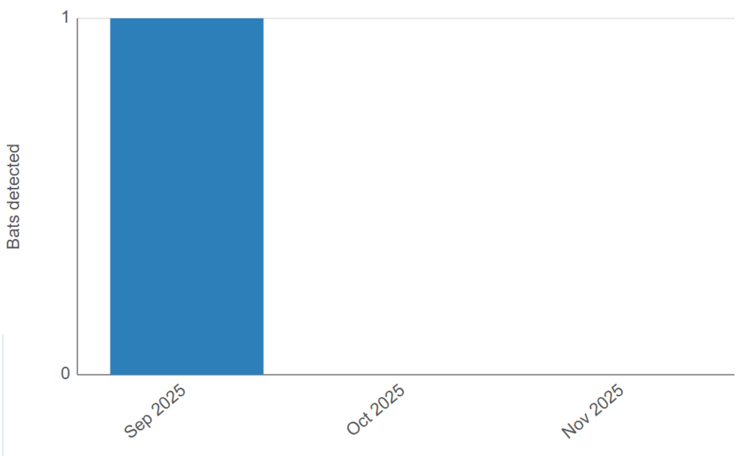
NRCAN-Discovery

Species Code	Total Detected
LACI	1
Noise	9,059
Total Bats	**1**



VES-NOC-RRS-JamesCook

Species Code	Total Detected
LABOMYLU	1
Noise	7,476
Total Bats	**1**



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